

Suppose the combustion reaction



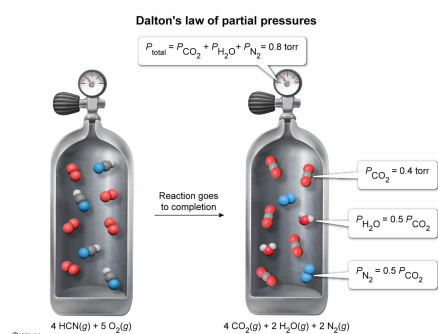
goes to completion in a sealed vessel. After the reaction, if the partial pressure of $\text{CO}_2(g)$ is 0.4 torr, the total pressure of the system is:

- ☐ A. 0.4 torr [4%]
✓ ☒ B. 0.8 torr [68%]
☐ C. 1.2 torr [13%]
☐ D. 1.7 torr [12%]

Correct

69% answered correctly

Explanation:



Dalton's law of partial pressures states that for a mixture of gases, the **total pressure** P_{total} is equal to the sum of the **partial pressures** P_i of each constituent gas at constant volume and temperature.

$$P_{\text{total}} = \sum P_i = P_1 + P_2 + P_3 + \dots + P_i$$

The partial pressure of each gas species in a mixture at a given temperature depends only on the relative number of moles of each species and not the identity of the gas. If the partial pressure of one gas species produced in a reaction is known, the other constituent gas species can be determined using the **stoichiometric coefficients** from a **balanced equation**.

The question states that after the combustion reaction



has reached completion (ie, the reactants are fully consumed), the resulting partial pressure of $\text{CO}_2(g)$ (P_{CO_2}) is 0.4 torr. According to the stoichiometric coefficients, the number of moles of $\text{H}_2\text{O}(g)$ and $\text{N}_2(g)$ formed are half that of $\text{CO}_2(g)$ formed. Consequently, the partial pressures of $\text{H}_2\text{O}(g)$ and $\text{N}_2(g)$ are equal to

$$P_{\text{H}_2\text{O}} = \frac{1}{2} P_{\text{CO}_2} = \frac{1}{2} \times 0.4 \text{ torr} = 0.2 \text{ torr}$$

$$P_{\text{N}_2} = \frac{1}{2} P_{\text{CO}_2} = \frac{1}{2} \times 0.4 \text{ torr} = 0.2 \text{ torr}$$

Therefore, the total pressure of the system is

$$P_{\text{total}} = P_{\text{CO}_2} + P_{\text{H}_2\text{O}} + P_{\text{N}_2} = 0.4 \text{ torr} + 0.2 \text{ torr} + 0.2 \text{ torr} = 0.8 \text{ torr}$$

(Choice A) A pressure of 0.4 torr is the partial pressure of $\text{CO}_2(g)$, not the *total* pressure.

(Choice C) A total pressure of 1.2 torr is calculated if it is incorrectly assumed that $P_{\text{CO}_2} = P_{\text{H}_2\text{O}} = P_{\text{N}_2}$.

(Choice D) If the partial pressures of the *reactants* are included in the total pressure (ie, the reaction has not gone to completion), an incorrect total pressure of 1.7 torr is obtained:

$$P_{\text{total}} = P_{\text{HCN}} + P_{\text{O}_2} + P_{\text{CO}_2} + P_{\text{H}_2\text{O}} + P_{\text{N}_2}$$

$$P_{\text{total}} = P_{\text{CO}_2} + \frac{5}{4}P_{\text{CO}_2} + P_{\text{CO}_2} + \frac{1}{2}P_{\text{CO}_2} + \frac{1}{2}P_{\text{CO}_2} = 1.7 \text{ torr}$$

Educational objective:

The total pressure exerted by a mixture of gases is equal to the sum of the partial pressures of each gas component.

The partial pressure of each gas species in a mixture depends only on the relative number of moles of each species and not the identity of the gas.

Time spent:0 sec

Subject:
General Chemistry

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System:
Thermodynamics, Kinetics & Gas Laws